Somayeh Ghaneameleh

MSc Environmental Engineering | Prospective PhD Researcher

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Research Profile

MSc graduate in Biodiversity Management & Conservation with strong expertise in molecular phylogenetics, phylogeography, and conservation genetics. Experienced in full field-to-lab molecular workflows including DNA extraction, PCR, gel electrophoresis, sequencing preparation, and multi-gene mitochondrial analyses. Skilled in Bayesian and Maximum Likelihood phylogenetic inference (BEAST, MrBayes, RAxML), population genetic analyses, and spatial interpretation using R and GIS tools.

Research Interests

- **Molecular phylogenetics and evolutionary history**
- **Phylogeography and population structure**
- **Conservation genetics and management units**
- **Climate change impacts on biodiversity**
- **Species distribution and ecological niche modelling**

Education

2017-2020. M.Sc. in Science and Environmental Engineering- Biodiversity Management and Conservation

- **Shahid Beheshti University, Tehran, Iran (www.sbu.ac.ir)** **GPA: (17.60/20)**

  Thesis: Phylogeography of Three-lined Lizard (*Lacerta media*) in Iran

  Supervisor: Dr. Faraham Ahmadzadeh (f_ahmadzade@sbu.ac.ir)

2010-2014. B.Sc. in Natural Resources Engineering- Environmental Science

- **Kurdistan University, Kurdistan, Iran (www.uok.ac.ir)** **GPA: (15.16/20)**

  Project: Explaining the vital components, geography and biological threats of *Neurergus derjugini*

  Supervisor: Dr. Hanyeh Ghaffari (h.ghaffari@uok.ac.ir)

Honors

Ranked 25th among >2500 participants in the Iranian National University Entrance Exam, Science and Environmental Engineering **(MSc) 2017.**

Publication

- **Somayeh Ghane-Ameleh**, Milad Khosravi, Reihaneh Saberi-Pirooz, Elham Ebrahimi, Marzieh Asadi Aghbolaghi, and Faraham Ahmadzadeh. **(2021)**. "Mid-Pleistocene Transition as a Trigger for Diversification in the Irano-Anatolian Region: Evidence Revealed by Phylogeography and Distribution Pattern of the Eastern Three-Lined Lizard." **Global Ecology and Conservation** https://doi.org/10.1016/j.gecco.2021.e01839
- Mohsen Ahmadi, Mahmoud-Reza Hemami, Mohammad Kaboli, **Somayeh Ghane-Ameleh** and Mansoureh Malekian. **(2024)**. "Conservation Biogeography of Mountain Vipers: A Phylogenetic Niche Modelling Approach." **Diversity and Distributions.** https://doi.org/10.1111/ddi.13955

Research Experience

Performed research under Dr. Faraham Ahmadzadeh in the Laboratory of the university, on the DNA extraction and Polymerase chain reactions (PCRs) using a Thermomixer, Microcentrifuge, Thermocycler (Bio-Rad), Safe-Red™, AutoDocument and Autoclave with the team working in the Laboratory.

Skills

- **Field Working and Laboratory Skills**

Sampling in the environment with team working in the field, Sterilization of devices with moist heat, removing portion of muscle tissue, DNA extraction of muscle tissue, DNA purification and PCR, Nanodrop Spectrometry, Electrophoresis (SDS- PAGE and Agarose), Visualization of amplified DNA fragments, prepare sequences for sequencing, Freezing of genetic samples.

- **Software**

R Programming, Geneious prime, MrModeltest, RAxML, SequenceMatrix, MrBayes, MEGA, DnaSP, Arlequin, TCS, FigTree, Tracer, PAUP, BEAST, ArcGIS, Microsoft Office Package.

- **Languages**

Kurdish (native) - Persian (native) - English (Fluent).

Work Experience

Environmental Scientists nec
Bina Azma Sepahan Company | Sep 2014 – Present | Full-time

- Conduct environmental sampling and laboratory analysis (air, water, soil, wastewater, food products).
- Perform data analysis and environmental risk assessment using statistical approaches.
- Contribute to pollution monitoring, wastewater treatment evaluation, and industrial environmental compliance strategies.

Certificate

Learn to Program Python: A 6-hour course delivered by Intersect Australia on Wednesday, August 31, and Thursday, September 1, 2022, at the University of Canberra**.**

Data Visualisation in R: A 3-hour course delivered by Intersect Australia on Friday, August 19, 2022 at Deakin University.

Data Manipulation in R: A 3-hour course delivered by Intersect Australia on Thursday, August 18, 2022 at Deakin University.

R Programming Software; Matris Pedia Training (MPT) Iran Representative, 2022.

Professional Activities

Attendance in t7th7st national conference on air pollution and sound management, Shahid Beheshti University, Tehran, Iran, (January. 28-29, 2019).

Member of the 7th **national conference on air pollution** and sound management, Shahid Beheshti University, Tehran, Iran, (January. 28-29, 2019).

Languages

Kurdish (native), Persian (native), English (fluent).

References

Dr. Faraham Ahmadzadeh Associate Professor of Shahid Beheshti University, Tehran, Iran, Institute (ESRI).

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Dr. Hanyeh Ghaffari Associate Professor of Kurdistan University, Kurdistan, Iran, Department of Environment, Faculty of Natural Resource.

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